Inequality Exercises: Cauchy, AM–GM, Vectors, and Optimization

N-20220811

§1.1 How to read this note

A compact inequality practice set is most useful when it reveals the standard inequality hidden inside each expression. The useful question is not "which formula should be memorized?", but rather "what structure does the expression already have?"

If there are fractions with variable denominators, Cauchy or Engel form is natural. If there are products such as abc, AM–GM is natural. If there are square roots of quadratic expressions, think of vector lengths and the triangle inequality. If the problem asks for a maximum or minimum under a quadratic constraint, expect equality cases in Cauchy–Schwarz.

§1.2 A fraction problem with a constraint

One problem asks to handle expressions of the form

$$\frac{4}{4-x^2}+\frac{9}{9-y^2}$$

under constraints such as $xy=-1$ and bounded intervals for x, y. A Cauchy-style lower bound gives:

$$\frac{4}{4-x^2}+\frac{9}{9-y^2} \geq \frac{(2+3)^2}{(4-x^2)+(9-y^2)}.$$

Then the constraint controls the denominator. The important point is that the numerator $4, 9$ are squares, so the expression is inviting the Cauchy inequality

$$\frac{a^2}{u}+\frac{b^2}{v} \geq \frac{(a+b)^2}{u+v}.$$

This is one of the most common contest moves: when a sum of fractions has square numerators, try to combine them by Cauchy rather than simplifying separately.

§1.3 A weighted square-root comparison

Another exercise defines

$$P=\sqrt{ab}+\sqrt{cd},$$

and

$$Q=\sqrt{ma+nc}\sqrt{\frac{b}{m}+\frac{d}{n}},$$

where the variables are positive. Expanding Q^2 gives

$$\begin{aligned} Q^2 &= (ma + nc)\left(\frac{b}{m} + \frac{d}{n}\right) \\ &= ab + cd + \frac{mad}{n} + \frac{ncb}{m}. \end{aligned}$$

By AM–GM,

$$\frac{mad}{n} + \frac{ncb}{m} \geq 2\sqrt{abcd}.$$

Hence

$$Q^2 \geq ab + cd + 2\sqrt{abcd} = (\sqrt{ab} + \sqrt{cd})^2 = P^2.$$

So $P \leq Q$, with equality exactly when

$$\frac{mad}{n} = \frac{ncb}{m}.$$

Strict inequality holds precisely when this equality condition fails.

§1.4 Maximizing a radical expression

The note then asks for the maximum of

$$f(x) = 2\sqrt{x-3} + \sqrt{5-x}, \qquad 3 \leq x \leq 5.$$

One way is calculus. We have

$$f'(x) = \frac{1}{\sqrt{x-3}} - \frac{1}{2\sqrt{5-x}}.$$

Setting $f'(x) = 0$ gives

$$2\sqrt{5-x} = \sqrt{x-3},$$

so

$$4(5-x) = x-3, \qquad x = \frac{23}{5}.$$

At this point,

$$f\left(\frac{23}{5}\right) = 2\sqrt{\frac{8}{5}} + \sqrt{\frac{2}{5}} = \sqrt{10}.$$

The same conclusion can be reached without derivatives, by Cauchy–Schwarz:

$$\left(2\sqrt{x-3} + \sqrt{5-x}\right)^2 \leq (4+1)((x-3) + (5-x)) = 10.$$

Equality holds when

$$\frac{2}{1} = \frac{\sqrt{x-3}}{\sqrt{5-x}},$$

which again gives $x = 23/5$. This non-calculus proof is more elegant because it explains the form of the answer.

§1.5 Equality in Cauchy–Schwarz

A vector problem gives

$$a^2+b^2+c^2=25, \qquad x^2+y^2+z^2=36, \qquad ax+by+cz=30.$$

By Cauchy–Schwarz,

$$(ax+by+cz)^2 \le (a^2+b^2+c^2)(x^2+y^2+z^2) = 25\cdot 36 = 900.$$

Since $ax+by+cz=30$, equality holds. Therefore

$$(a,b,c)=k(x,y,z).$$

Comparing norms gives

$$k=\frac{5}{6}.$$

Thus

$$\frac{a+b+c}{x+y+z}=\frac{5}{6}.$$

This is an excellent example of why equality cases matter. The inequality is not just bounding a number; it tells us the exact geometry of the vectors.

§1.6 Triangle-angle inequalities

Let A,B,C be the angles of a triangle. Then

$$A+B+C=\pi.$$

By Cauchy–Schwarz,

$$(A+B+C)\left(\frac{1}{A}+\frac{1}{B}+\frac{1}{C}\right) \ge (1+1+1)^2 = 9.$$

Hence

$$\frac{1}{A}+\frac{1}{B}+\frac{1}{C} \ge \frac{9}{\pi}.$$

Similarly,

$$A^2+B^2+C^2 \ge \frac{(A+B+C)^2}{3} = \frac{\pi^2}{3}.$$

Both equalities occur only for

$$A=B=C=\frac{\pi}{3}.$$

Symmetric triangle inequalities often attain equality at the equilateral triangle.

§1.7 Two spheres and a linear equation

Another problem asks to solve a system of two sphere equations. The trick is to subtract them. If two equations are

$$(x-a_1)^2+(y-b_1)^2+(z-c_1)^2=R_1^2,$$

$$(x-a_2)^2+(y-b_2)^2+(z-c_2)^2=R_2^2,$$

subtracting cancels $x^2 + y^2 + z^2$ and leaves a plane. The intersection of two spheres is therefore a circle, a point, or empty; algebraically it becomes a sphere-plane problem.

Cauchy–Schwarz bounds a linear expression such as

$$-8x + 6y - 24z.$$

The equality condition determines the direction of the vector (x, y, z). Again the method is geometric: a linear function is maximized on a sphere in the direction of its coefficient vector.

§1.8 A product inequality

If

$$(1+a)(1+b)(1+c) = 8, \qquad a, b, c > 0,$$

then one obtains

$$abc \le 1.$$

Indeed,

$$\begin{aligned} 8 &= 1 + a + b + c + ab + bc + ca + abc \\ &\ge 1 + 3\sqrt[3]{abc} + 3\sqrt[3]{(ab)(bc)(ca)} + abc \\ &= 1 + 3t + 3t^2 + t^3 = (1+t)^3, \end{aligned}$$

where $t = \sqrt[3]{abc}$. Thus $1 + t \le 2$, so $t \le 1$, and therefore $abc \le 1$.

§1.9 Vector-style radical inequality

The pages also contain an inequality of the form

$$\sqrt{x^2 + xy + y^2} + \sqrt{x^2 + xz + z^2} \ge \sqrt{y^2 + yz + z^2}.$$

This is best understood as a disguised triangle inequality. One rewrites each quadratic as a squared Euclidean norm of a two-dimensional vector. For example,

$$x^2 + xy + y^2 = \left(x + \frac{y}{2}\right)^2 + \frac{3}{4}y^2.$$

Once each radical is interpreted as a vector length, the inequality becomes

$$\|u\| + \|v\| \ge \|u + v\|.$$

This is a typical high-level move: turn algebraic square roots into geometry.

§1.10 The deeper habit

The note is not merely a list of inequality tricks. It repeatedly uses the same three ideas: Cauchy detects equality of vectors, AM–GM detects equality of positive quantities, and radical inequalities often hide triangle inequalities. The more mature way to remember the sheet is therefore:

$$\text{fractions} \to \text{Cauchy}, \qquad \text{products} \to \text{AM–GM}, \qquad \text{radicals} \to \text{norms}.$$

§1.11 Equality cases as structural information

In inequality problems, equality conditions are not afterthoughts. They often reveal the correct substitution or normalization. For Cauchy–Schwarz,

$$\left(\sum_i a_i b_i\right)^2 = \left(\sum_i a_i^2\right)\left(\sum_i b_i^2\right)$$

holds exactly when a and b are proportional. If a proposed solution cannot recover the equality case, it is probably not using the natural structure.

§1.12 Optimization on constrained domains

Many radical or fraction inequalities are optimization problems on a constrained set. A constraint such as $x+y+z=1$ suggests compactness if variables are nonnegative. Then a maximum exists, and one can use convexity, Cauchy, or Lagrange multipliers to locate it.

This separates two tasks: first prove existence or reduce the domain; then identify the inequality that gives the optimum.

§1.13 Vectors and geometric inequalities

Vector methods convert algebraic inequalities into length comparisons. If an expression can be written as a dot product, Cauchy applies. If it is a sum of lengths, triangle inequality or Minkowski applies.

The power of vectors is that equality becomes a geometric condition: parallel vectors, collinearity, or degenerate triangles.

§1.14 Equality cases as geometric diagnostics

The exercise sheet is about choosing the right geometry. Cauchy handles dot products, AM–GM handles balanced positive products, convexity handles averages, and Lagrange multipliers handle smooth constrained extrema. Equality cases are the map showing which method is natural.

§1.15 Equality cases as geometry

The original inequality exercises repeatedly ask when equality holds. This is not a side detail; equality describes the geometry of the inequality. In Cauchy–Schwarz,

$$\left(\sum a_i b_i\right)^2 \le \left(\sum a_i^2\right)\left(\sum b_i^2\right),$$

equality holds exactly when the vectors (a_i) and (b_i) are linearly dependent. In AM–GM,

$$\frac{x_1+\cdots+x_n}{n} \ge \sqrt[n]{x_1\cdots x_n},$$

equality holds when all variables are equal. These equality sets often reveal the hidden symmetry of a problem.

§1.16 Lagrange multipliers and inequalities

Many inequality exercises are optimization problems with constraints. If one wants to extremize $f(x_1, \ldots, x_n)$ under $g(x_1, \ldots, x_n) = 0$, the smooth interior candidates satisfy

$$\nabla f = \lambda \nabla g.$$

For example, optimizing a quadratic form under $\|x\| = 1$ leads to

$$Ax = \lambda x,$$

so eigenvalues are extrema of quadratic forms on the unit sphere. This is the Rayleigh quotient principle:

$$\lambda_{\min} \leq \frac{x^T Ax}{x^T x} \leq \lambda_{\max}$$

for symmetric A.

§1.17 Semidefinite viewpoint

Some inequalities are equivalent to positive semidefiniteness. To prove

$$ax^2 + 2bxy + cy^2 \geq 0$$

for all x, y, one checks

$$\begin{pmatrix} a & b \\ b & c \end{pmatrix} \succeq 0.$$

That means

$$a \geq 0, \qquad ac - b^2 \geq 0.$$

This connects elementary quadratic inequalities to semidefinite programming, where one optimizes linear functions over the cone of positive semidefinite matrices.

§1.18 Classical inequalities encode convex optimization

The contest techniques in the source—Cauchy, AM–GM, substitutions, and equality checking—should be read as finite-dimensional optimization. The advanced language does not replace the computations; it explains why the same few patterns recur: projection, convexity, eigenvalues, and positive semidefinite forms.